

Lab report # 11

1. Objective

The objective of this lab is to:

- Understand C-Strings in C++
 - Learn declaration and initialization of C-Strings
 - Understand string objects in C++
 - Learn commonly used string handling functions
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2. Theory

2.1 C++ Strings

A string is a collection of characters. In C++, there are two main types of strings:

1. C-Strings (C-style strings)
2. String class objects (standard C++ string library)

2.2 C-String

A C-string is a character array that ends with a null character `\0`.

3. It is similar to arrays in C++
 4. It is always terminated by `\0`
 5. ASCII value of `\0` is 0
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Example 1: Basic C-String Initialization

```
#include <iostream>
using namespace std;

int main()
{
    char myString[] = "Hello"; // yahan string initialize ki ja rahi hai
```

```
    cout << myString << endl; // yahan string display ki ja rahi hai

    return 0; // program yahan terminate hota hai
}
```

Example 2: Fixed Size C-String

```
#include <iostream>
using namespace std;

int main()
{
    char myString[6] = "Hello"; // yahan fixed size array mein string store
    ki gayi hai

    cout << myString << endl; // yahan output display kiya ja raha hai

    return 0;
}
```

Example 3: Character Array Method

```
#include <iostream>
using namespace std;

int main()
{
    char myString[] = {'H','e','l','l','o','\0'}; // yahan manually null
    character add kiya gaya hai

    cout << myString << endl; // string output show ki ja rahi hai

    return 0;
}
```

3. String Input Problem (cin)

```
#include <iostream>
using namespace std;

int main()
{
    char str[100]; // yahan string store karne ke liye array banaya gaya hai

    cout << "Enter a string: "; // user se input manga ja raha hai
    cin >> str; // yahan input sirf pehle space tak read hota hai

    cout << "You entered: " << str << endl; // output display ki ja rahi hai
}
```

```
    return 0;
}
```

4. Full Line Input Using cin.get()

```
#include <iostream>
using namespace std;

int main()
{
    char str[100]; // yahan character array define ki gayi hai

    cout << "Enter a string: "; // user ko input dene ka message
    cin.get(str, 100); // yahan poori line read ki ja rahi hai (spaces ke
    sath)

    cout << "You entered: " << str << endl; // complete string display ki ja
    rahi hai

    return 0;
}
```

5. String Object (Modern C++)

```
#include <iostream>
#include <string>
using namespace std;

int main()
{
    string str; // yahan string object declare kiya gaya hai

    cout << "Enter a string: "; // user se input manga ja raha hai
    getline(cin, str); // yahan poori line input li ja rahi hai

    cout << "You entered: " << str << endl; // string output display ki ja
    rahi hai

    return 0;
}
```

6. String Functions Example

```
#include <iostream>
#include <cstring>
using namespace std;

int main()
```

```
{
    char s1[50] = "Hello"; // pehli string initialize ki gayi hai
    char s2[50] = "World"; // doosri string initialize ki gayi hai

    cout << "Length of s1: " << strlen(s1) << endl; // yahan string ki length
    nikal rahe hain

    strcat(s1, s2); // yahan s2 ko s1 ke sath jora ja raha hai

    cout << "After concatenation: " << s1 << endl; // final string display ki
    ja rahi hai

    return 0;
}
```

7. Conclusion

In this lab, we learned:

- How C-Strings work using character arrays
- Difference between `cin`, `cin.get()`, and `getline()`
- Usage of string class in C++
- Basic string manipulation functions